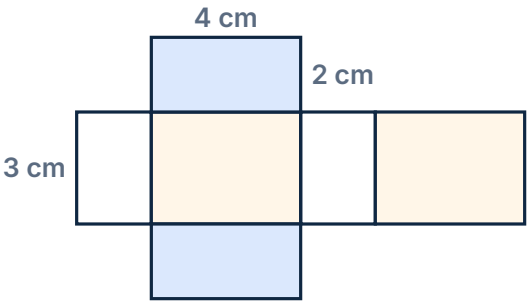
# Surface Area and Nets

Open the box flat. Add the areas of all the faces.

### A box 4 by 2 by 3, opened flat (a net)




6 faces, in 3 matching pairs.

top and bottom: 2 × 8 = 16

front and back: 2 × 12 = 24

two sides: 2 × 6 = 12

**16 + 24 + 12 = 52 sq cm**

**Surface area is square units (covering). Volume is cubic units (filling).**

**Your turn!**

cube, edge 5 cm: 6 × 25 = 150 sq cm

**For teachers**

Goal: surface area from nets of boxes and square pyramids (6.G.4). Say “Which faces match? Find one, double it.” Surface area is covering (sq units); volume is filling (cubic units).

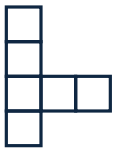
# Surface Area and Nets: Practice

Find each face. Matching faces come in pairs.

## Part 1 Does it fold into a cube? Imagine folding it up. Circle yes or no.

1) 

(yes) no

2) 

yes (no)

3) 

(yes) no

4) 

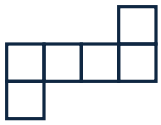
yes (no)

5) 

yes (no)

6) 

(yes) no

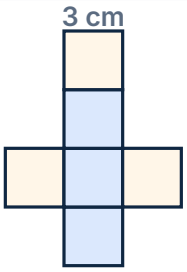
7) 

(yes) no

8) 

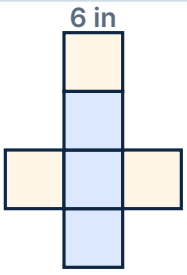
yes (no)

## Part 2 Cubes Six faces, all the same.

9) 


one face: 9

6 × 9 = 54 sq cm

10) 

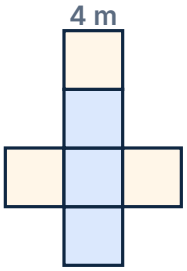

one face: 36

6 × 36 = 216 sq in

## Surface Area and Nets: Practice (continued)

### Part 2 Cubes (continued)

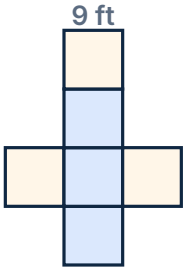
11)




one face: 16

6 × 16 = 96 sq m

12)

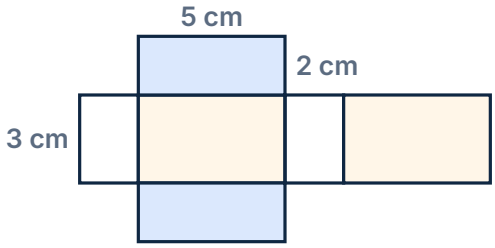



one face: 81

6 × 81 = 486 sq ft

### Part 3 Boxes Three pairs of faces.

13)



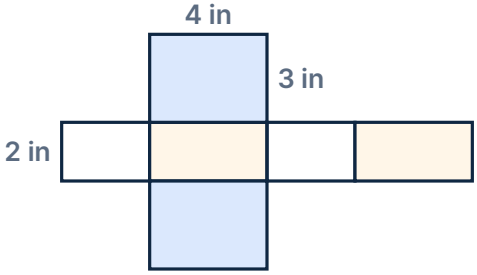

top and bottom: 2 × 10 = 20

front and back: 2 × 15 = 30

two sides: 2 × 6 = 12

surface area = 62 sq cm

14)




top and bottom: 2 × 12 = 24

front and back: 2 × 8 = 16

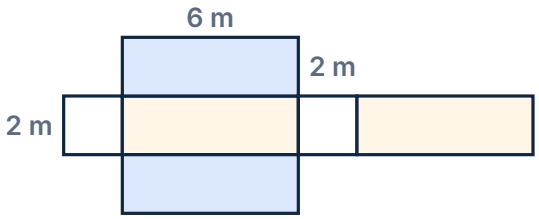
two sides: 2 × 6 = 12

surface area = 52 sq in

## Surface Area and Nets: Practice (continued)

### Part 3 Boxes (continued)

15)



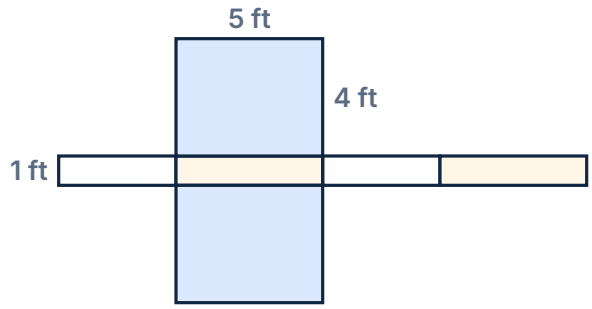

top and bottom: 2 × 12 = 24

front and back: 2 × 12 = 24

two sides: 2 × 4 = 8

surface area = 56 sq m

16)



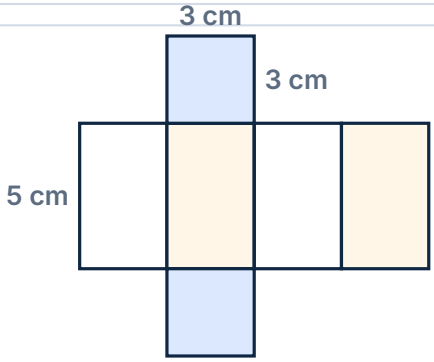

top and bottom: 2 × 20 = 40

front and back: 2 × 5 = 10

two sides: 2 × 4 = 8

surface area = 58 sq ft

17)



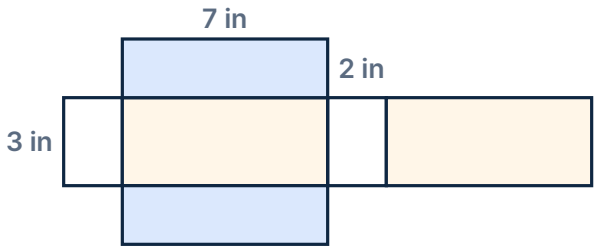

top and bottom: 2 × 9 = 18

front and back: 2 × 15 = 30

two sides: 2 × 15 = 30

surface area = 78 sq cm

18)




top and bottom: 2 × 14 = 28

front and back: 2 × 21 = 42

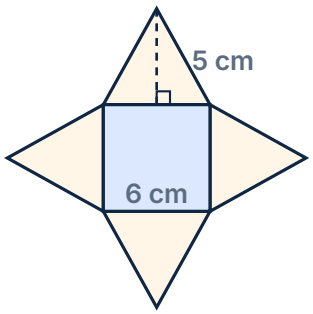
two sides: 2 × 6 = 12

surface area = 82 sq in

## Surface Area and Nets: Practice (continued)

**Part 4** Square pyramids The square, then 4 triangles.

19)

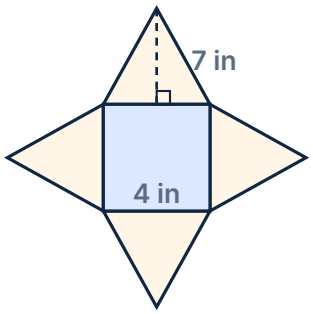



square: 36

4 × 15 = 60

surface area = 96 sq cm

20)

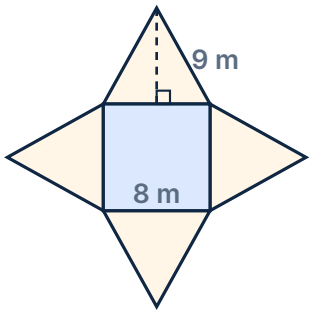



square: 16

4 × 14 = 56

surface area = 72 sq in

21)

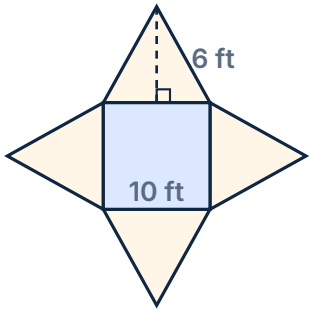



square: 64

4 × 36 = 144

surface area = 208 sq m

22)




square: 100

4 × 30 = 120

surface area = 220 sq ft

## Surface Area and Nets: Practice (continued)

**Part 5** Boxes No help this time.

23) box 6 × 4 × 3 cm: surface area = 108 sq cm

24) box 5 × 5 × 2 in: surface area = 90 sq in

25) box 8 × 3 × 2 m: surface area = 92 sq m

26) box 7 × 4 × 5 ft: surface area = 166 sq ft

27) box 10 × 2 × 3 cm: surface area = 112 sq cm

28) box 9 × 6 × 1 m: surface area = 138 sq m

29) box 4 × 4 × 7 in: surface area = 144 sq in

30) box 12 × 3 × 2 cm: surface area = 132 sq cm

**Part 6** Cubes No help this time.

31) cube, edge 7 cm: surface area = 294 sq cm

32) cube, edge 2 m: surface area = 24 sq m

33) cube, edge 8 in: surface area = 384 sq in

## Surface Area and Nets: Practice (continued)

### Part 6 Cubes (continued)

34) cube, edge 10 ft: surface area = 600 sq ft

35) cube, edge 11 cm: surface area = 726 sq cm

36) cube, edge 12 in: surface area = 864 sq in

### Part 7 More boxes

37) box 12 × 9 × 3 ft: surface area = 342 sq ft

38) box 11 × 7 × 1 cm: surface area = 190 sq cm

39) box 15 × 8 × 8 in: surface area = 608 sq in

40) box 11 × 4 × 8 ft: surface area = 328 sq ft

41) box 13 × 7 × 7 m: surface area = 462 sq m

42) box 10 × 7 × 8 m: surface area = 412 sq m

43) box 7 × 2 × 6 cm: surface area = 136 sq cm

44) box 6 × 2 × 2 ft: surface area = 56 sq ft

## Surface Area and Nets: Practice (continued)

### Part 7 More boxes (continued)

45) box 12 × 7 × 7 m: surface area = 434 sq m

46) box 13 × 10 × 1 cm: surface area = 306 sq cm

47) box 5 × 3 × 3 cm: surface area = 78 sq cm

48) box 10 × 4 × 6 m: surface area = 248 sq m

49) box 13 × 3 × 1 m: surface area = 110 sq m

50) box 11 × 9 × 5 in: surface area = 398 sq in

51) box 9 × 6 × 2 ft: surface area = 168 sq ft

52) box 14 × 7 × 8 cm: surface area = 532 sq cm

53) box 12 × 5 × 4 cm: surface area = 256 sq cm

54) box 14 × 5 × 2 cm: surface area = 216 sq cm

## Surface Area and Nets: Practice (continued)

### Part 8 More cubes

55) cube, edge 8 ft: surface area = 384 sq ft

56) cube, edge 12 cm: surface area = 864 sq cm

57) cube, edge 17 cm: surface area = 1,734 sq cm

58) cube, edge 4 ft: surface area = 96 sq ft

59) cube, edge 19 m: surface area = 2,166 sq m

60) cube, edge 20 ft: surface area = 2,400 sq ft

61) cube, edge 14 m: surface area = 1,176 sq m

62) cube, edge 3 ft: surface area = 54 sq ft

## Surface Area and Nets: Practice (continued)

### Part 9 Remember? Problems from earlier packets.

63) $2\frac{1}{2} \times 2 \times 6 =$ **30**

64) $5x = 85$: $x =$ **17**

65) $2.6 + 13.75 =$ **16.35**

66) $-9$ **<** $-4$

67) when $n = 6$: $3n + 11 =$ **29**

68) 18 out of 20 = **90** %

### Part 10 Word problems Draw the net. Add the faces.

69) A cube box has edges of 4 in. How much paper covers it all?

Number sentence **6 × 16 = 96**

Answer **96** sq in

Draw or show your work

70) A box is 5 cm by 3 cm by 2 cm. What is its surface area?

Number sentence **30 + 20 + 12 = 62**

Answer **62** sq cm

Draw or show your work

71) A gift box is 10 in by 6 in by 4 in. How much wrapping paper covers it?

Number sentence **120 + 80 + 48 = 248**

Answer **248** sq in

Draw or show your work

## Surface Area and Nets: Practice (continued)

### Part 10 Word problems (continued)

72) A tent has 4 triangle sides, each with base 3 m and height 4 m. How much cloth for the sides?

Number sentence 3 × 4 ÷ 2 = 6, 4 × 6 = 24

Answer 24 sq m

Draw or show your work

73) Challenge: A box 6 × 4 × 2 ft has no top. How much wood makes it?

Number sentence 96 − 24 = 72

Answer 72 sq ft

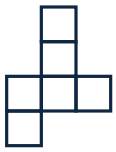
Draw or show your work

# Check-In: Surface Area and Nets

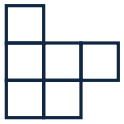
Show what you know. No help this time.

Mastery: 5 of 6 (80%)

## Part 1 Does it fold into a cube? Circle yes or no.

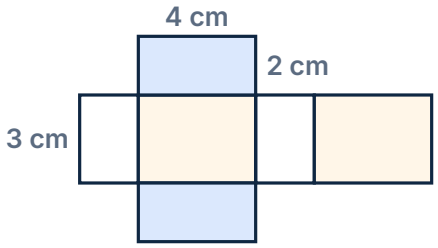
1) 

(yes) no

2) 

yes (no)

## Part 2 Boxes

3) 


top and bottom: 2 × 8 = 16

front and back: 2 × 12 = 24

two sides: 2 × 6 = 12

surface area = 52 sq cm

## Part 3 Find the surface area

4) cube, edge 5 in: surface area = 150 sq in

5) cube, edge 9 m: surface area = 486 sq m

6) box 8 × 2 × 3 in: surface area = 92 sq in